



Comprehensive Examination (M) – Computer Programming (CS F111)

First Semester: 2024-25

Department of Computer Science and Information Systems

Name and ID: _____

Important Information:

- Write your name and ID legibly on the question paper.
- Please check the completeness of the exam booklet (**8 double-sided pages**)
- Use only **blue** or **black** pen to answer the questions; Pencils, green and red pens are not allowed!
- Working time is **60 minutes** for **2 questions** of **30 points**!
- On any attempt to **cheat**: the exam will be assessed with a **0** and **you would be barred from makeup**.
- Permitted writing aids: T1 and printed slides + lab sheets — **open book exam**!

1. The following code snippet implements sorting the values of nodes of a singly [15 pts]
linked list. Sample execution is shown below. Fill in the blanks with appropriate
code segment(s).

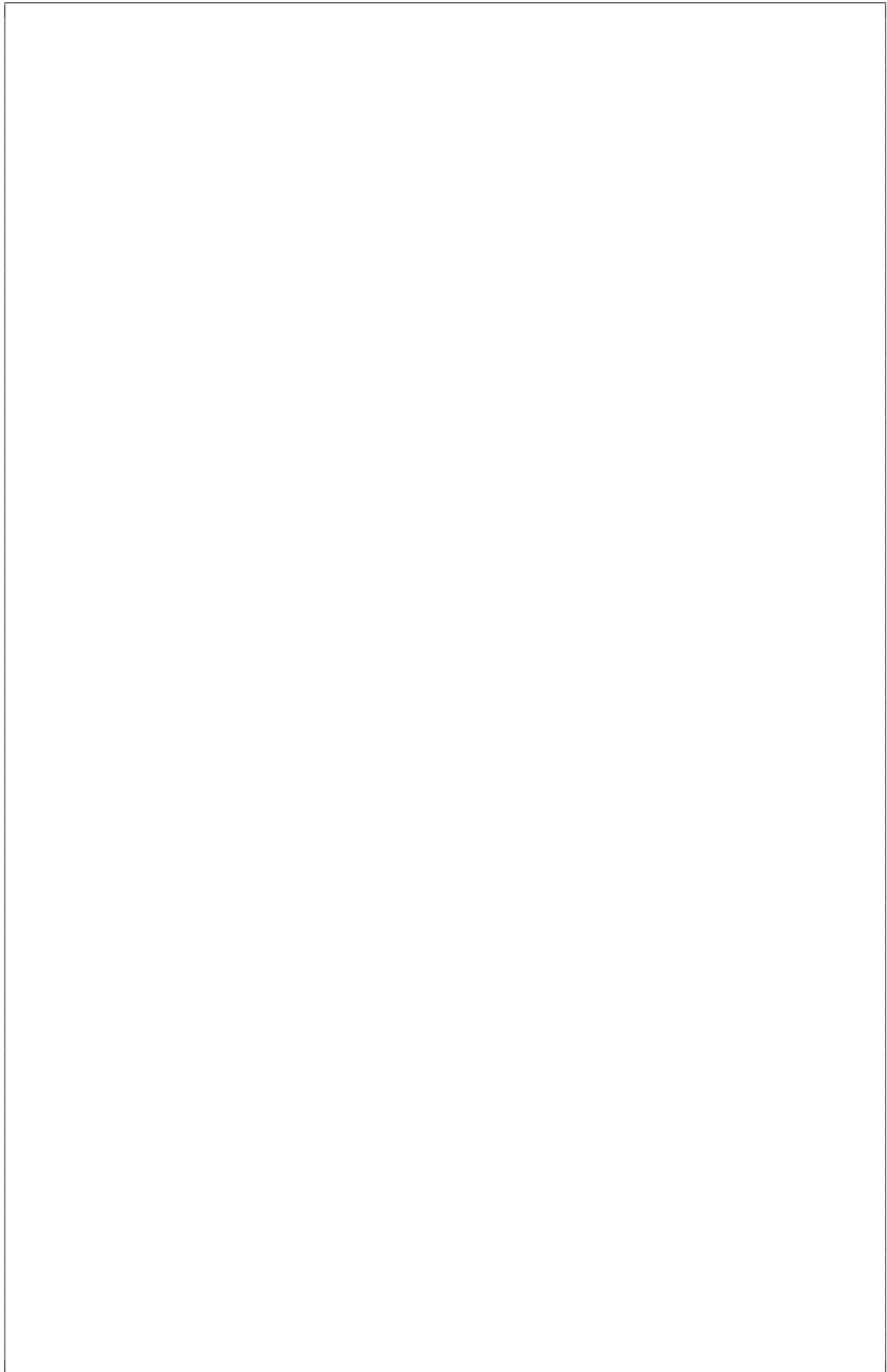
Question 1: 15 pts

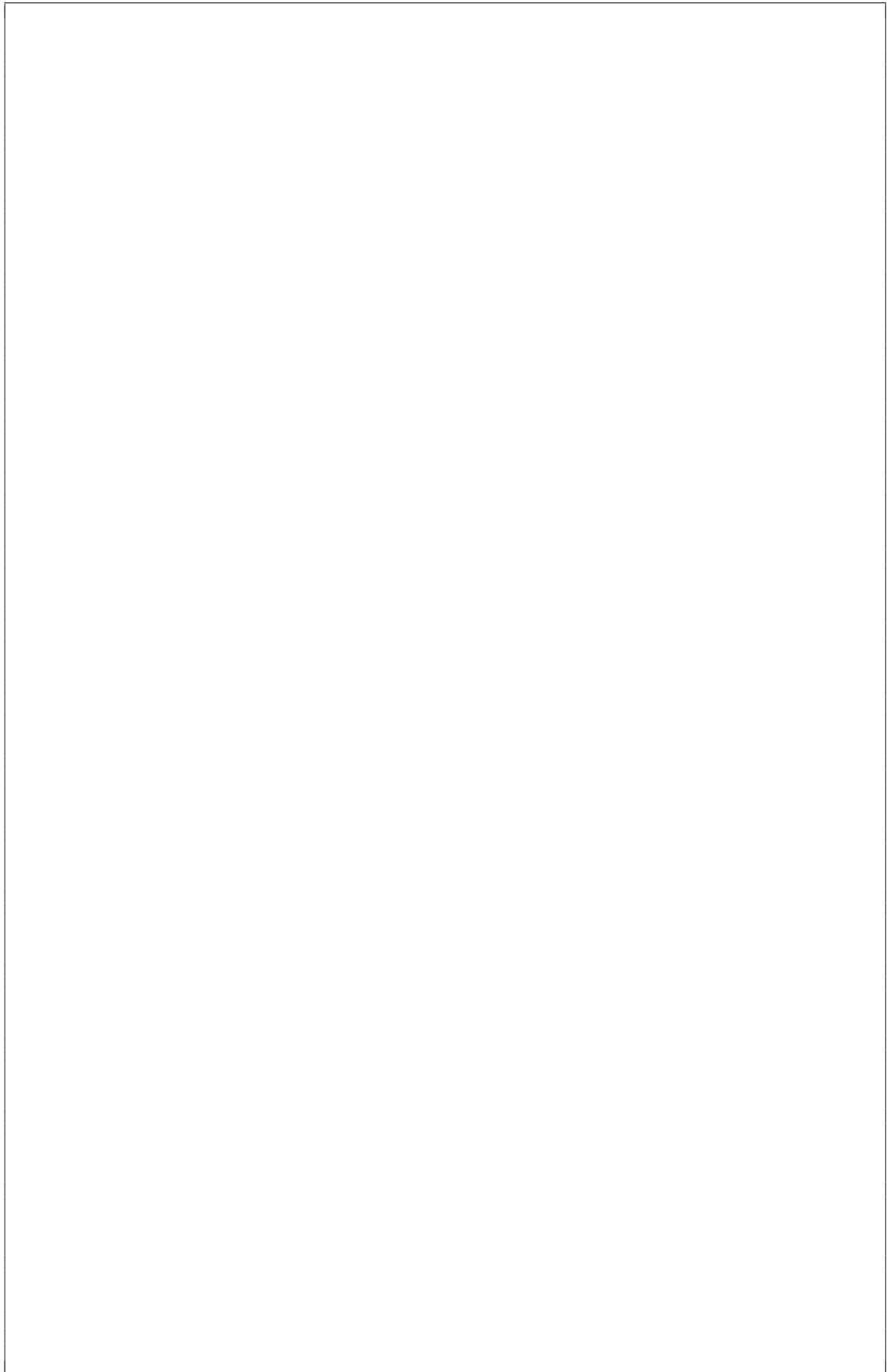
Expected output

```
enter the number of nodes:
6
enter the no: 33 66 11 77 44 22
the number are:
22 44 77 11 66 33
the sorted number are:
11 22 33 44 66 77
```

```
1      #include<stdio.h>
2      #include<malloc.h>
3      typedef struct node NODE;
4      struct node
5      {
6          int info;
7          struct node *ptr;
8      } ;
9      NODE *st1, *st2;
10     void swap(NODE *st1, NODE *st2 )
11     {
12         NODE *p;
13         int temp;
14         p=st1;
```

```
15         st2= p->ptr;
16         while(st2!=NULL)
17         {
18             -----
19             /* write your code for swapping based on
                comparison */ 10 Marks
20             -----
21         }
22     }
23
24     void show( NODE *st1, int m)
25     {
26         NODE *q;
27         int i;
28         q=st1;
29         for(i=1;i<=m;i++)
30         {
31             printf(" %d ",-----);    0.5 Mark
32             -----;                    0.5 Mark
33         }
34     }
35
36     int main()
37     {
38         int i,j,n;
39         NODE *p;
40         printf("\nenter the no. of nodes:\n");
41         scanf("%d",&n);
42         st1= st2=NULL;
43         printf("enter the numbers:\t");
44         for(i=1;i<=n;i++)
45         {
46             -----
47             /* write your code to create nodes using DMA and
                make a linked list, say, st1 */ 4 Marks
48             -----
49         }
50         printf("\n the numbers are:\n");
51         show(st1,n);
52         for(i=1;i<=n;i++)
53             swap(st1, st2);
54
55         printf("\n the sorted numbers are:\n");
56         show(st1,n);
57         return 0;
58     }
```





2. Program Outputs

Question 2: 15 pts

(a) Write the output of the following code snippet:

[06 pts]

```
1      #include <stdio.h>
2      int ans=0;
3      int func(int n, int list1[], int list2[])
4      {
5          static int iter = 3;
6          for (int i=0; i<n; i++) {
7              list1[i] = ++list1[i]+list2[i]++;
8              ans = ans + list2[i]+n;
9          }
10
11         if(iter==0)
12             return ans;
13         else
14             printf("Ans=%d\n", ans);
15
16         func(--iter,list1,list2);
17     }
18
19     int main() {
20         int array[3]= {'A','a','b'};
21         int count[3]= {1,2,3};
22         func(3,array,count);
23         printf("Array0=%d Array1=%d Array2=%d\n", array[0],array[1],array[2]);
24         return 0;
25     }
```

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(b) Write the output of the following code snippet:

[04 pts]

```
1      #include <stdio.h>
2      int val=1;
3      int func1(){
4          static int val=2;
5          return ++val + val++;
6      }
7      int func2(){
8          static int val=3;
9          func1();
```

```

10         return ++val + ++val;
11     }
12
13     int main(){
14         int value;
15         value = val++ +func1()*func2();
16         printf("Value1= %d",value);
17         value+= value + value *func2();
18         printf("\nValue2= %d",value);
19         return 0;
20     }

```

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(c) Write the output of the following code snippet:

[05 pts]

```

1         #include <stdio.h>
2         int func1(int);
3         int func2(int);
4
5         int main(){
6             int res1,res2;
7             printf("%d\n",func2(res1=func1(res2=func1
8                 (20))));
9             printf("Res1=%d Res2=%d\n", res1, res2);
10            return 0;
11        }
12        int func1(int a){
13            static int b=5;
14            if (b<9)
15            {
16                func1(++a/2+b++)+func2(++a/2+b++);
17            }
18            else
19                return func2(++a/2+b++);
20        }
21        int func2(int a){
22            static int b=3;
23            if (b<9)
24            {
25                func1(++a/2+b++)+func2(a/2+b++);
26            }
27            else

```

```
28         return a/2+b++;  
29     }
```

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(Don't let fear get the better of you. All the best!)

Space for Rough Work